

Architecture Reference

Architecture Assumptions

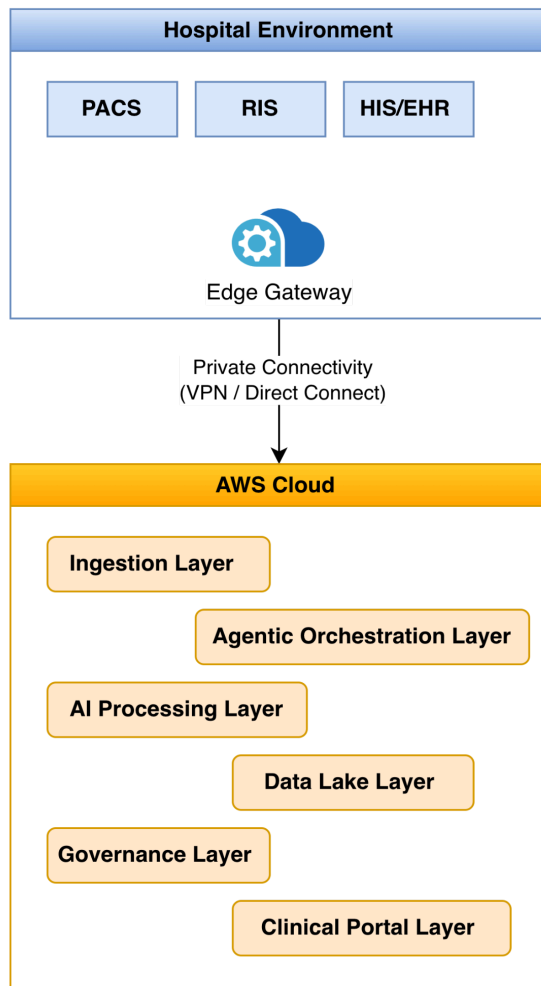
Let's define realistic assumptions.

Healthcare Context

- Hospital owns DICOM data.
- PACS remains on-premises.
- GDPR applies.
- Clinicians access hospital network.
- AI platform is operated in AWS.
- Some hospitals may prohibit raw DICOM storage in cloud.
- Human validation is mandatory.

Therefore: This is fundamentally a Hybrid Architecture.

High-Level AWS Architecture



The core intelligence of the platform is not implemented through LLM agents. The platform primarily relies on deterministic orchestration (Step Functions), specialized AI services (SageMaker/EKS), and governed workflow automation. Generative AI capabilities are introduced only where natural language reasoning provides measurable value, such as clinical explanation, reporting assistance, and knowledge retrieval.

Hospital Environment (Edge) Overview

The Hospital Edge Layer is a containerized platform deployed within the hospital network that acts as the local execution environment for all clinical workflows involving medical imaging and AI-assisted decision support.

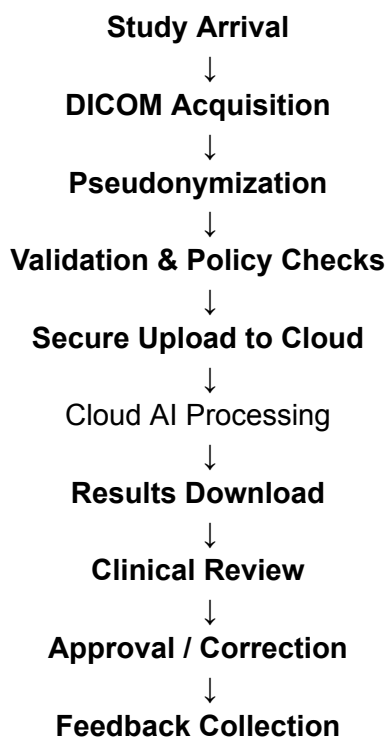
Its primary responsibilities are:

- Integration with hospital systems (PACS, RIS, HIS/EHR).
- DICOM acquisition and metadata extraction.
- Patient data pseudonymization and privacy enforcement.

- Validation of studies before cloud processing.
- Local workflow orchestration and state management.
- Secure synchronization with the Cloud AI Platform.
- Clinical review and validation of AI-generated results.
- Feedback collection for continuous model improvement.

The Edge Layer is not responsible for heavy AI computation. AI inference services such as anatomical segmentation, anomaly detection, and model training are executed in the Cloud AI Domain, where scalable GPU resources and centralized MLOps capabilities are available.

A typical workflow is:



From an architectural perspective, the Edge Layer acts as the hospital's **Clinical Control Plane**, while the Cloud Platform acts as the AI Computation and Learning Plane.

The preferred deployment model for CIOP is a containerized appliance based on **K3s**, hosting independently deployable services such as **Orthanc** (DICOM integration), **Keycloak** (identity management), **PostgreSQL** (metadata storage), **MinIO** (local object storage), Temporal (workflow orchestration), **Clinical UI services**, and a **Synchronization Engine**.

This approach provides portability across multiple healthcare facilities, simplified upgrades, operational consistency, offline resilience, and centralized lifecycle management while preserving compliance with healthcare privacy and governance requirements.

AWS Cloud Environment Overview

The AWS Cloud Environment represents the Intelligence and Computation Domain of the CIIO platform. Its primary purpose is to provide scalable AI processing, orchestration, data governance, and continuous learning capabilities while keeping clinical interaction and privacy-sensitive operations within the Hospital Edge Layer.

The Cloud Platform receives pseudonymized imaging studies and metadata from multiple healthcare facilities and processes them through a set of specialized AI and workflow services.

Its main responsibilities are:

- Secure ingestion of medical imaging studies and metadata.
- Centralized storage of imaging and derived datasets.
- AI-powered image segmentation and analysis.
- 3D anatomical reconstruction and model generation.
- Agentic orchestration of complex AI workflows.
- Human-in-the-Loop feedback processing.
- Model training, evaluation, and deployment.
- Monitoring, governance, and auditability.
- Multi-hospital scalability and resource sharing.

The AWS Cloud Environment can be logically organized into the following layers:

Ingestion Layer

Responsible for receiving data from hospital Edge platforms.

Typical services:

- Amazon S3
- Amazon API Gateway
- AWS Transfer Family (optional)
- Amazon EventBridge

Responsibilities:

- Secure upload
- Validation
- Event generation
- Routing

Data Management Layer

Central repository for all pseudonymized clinical data.

Typical services:

- Amazon S3 Data Lake
- Amazon DynamoDB
- AWS Glue Data Catalog

Responsibilities:

- Raw data storage
- Curated datasets
- Metadata management
- Data lineage

AI Processing Layer

Executes computationally intensive AI workloads.

Typical services:

- Amazon SageMaker
- Amazon EKS (GPU)
- AWS Batch

Responsibilities:

- Image segmentation
- Anomaly detection
- Inference execution
- Model hosting

3D Reconstruction Layer

Generates anatomical 3D models from segmented studies.

Typical services:

- Amazon EKS GPU Clusters
- AWS Batch

Responsibilities:

- Mesh generation
- Surface reconstruction
- FBX/OBJ generation

- Quality checks

Agentic Orchestration Layer

Coordinates complex workflows involving AI services, business rules, and human validation processes.

Typical services:

- AWS Step Functions
- Amazon EventBridge
- AWS Lambda
- Amazon Bedrock Agents (future evolution)

Responsibilities:

- Workflow coordination
- Decision routing
- Retry management
- Process automation

Human Feedback Intelligence Layer

Transforms clinical validation activities into structured learning assets.

Typical services:

- DynamoDB
- S3
- Lambda
- SageMaker Processing Jobs

Responsibilities:

- Feedback aggregation
- Correction analysis
- Dataset enrichment
- Active Learning support

MLOps & Continuous Learning Layer

Manages the complete AI model lifecycle.

Typical services:

- Amazon SageMaker Pipelines
- SageMaker Model Registry
- SageMaker Experiments

Responsibilities:

- Training
- Evaluation
- Model versioning
- Deployment governance

Security, Compliance & Governance Layer

Cross-cutting layer spanning the entire platform.

Typical services:

- IAM
- KMS
- CloudTrail
- AWS Config
- Security Hub

Responsibilities:

- Access control
- Encryption
- Auditing
- Compliance monitoring

Monitoring & Operations Layer

Provides operational visibility and reliability.

Typical services:

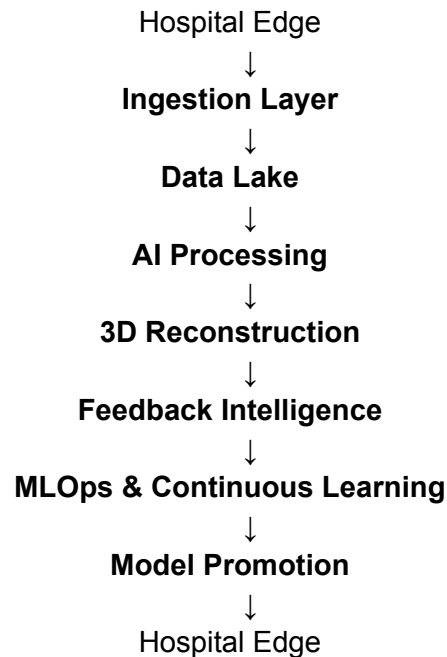
- CloudWatch
- X-Ray
- OpenSearch
- SNS

Responsibilities:

- Observability
- Alerting

- Performance monitoring
- Capacity management

High-Level Processing Flow



From an architectural perspective, the AWS Cloud Environment functions as a **centralized AI platform** serving multiple healthcare facilities. It provides elastic compute resources, centralized governance, reusable AI services, and continuous learning capabilities while allowing hospitals to maintain local control over clinical workflows, user interactions, and sensitive identity information.

A key design principle is that deterministic services (security, governance, workflow execution, compliance controls) remain separate from AI reasoning components, ensuring that clinical safety, traceability, and regulatory requirements are preserved even as Agentic AI capabilities evolve over time.